Teacher Wu Kuan | Exam at a Glance #28 — Exponential Decay & Solving with Logs

English lecture notes · YouTube Short 6cm3Y5WvNnQ

1. The trap question

Each filter keeps 80% of the light. How many filters to get below 10%?

2. The method

- After n filters: $0.8^n < 0.1$.
- Take log: $n \cdot \log 0.8 < -1$. $\log 0.8 = 3\log 2 - 1 \approx -0.0969$ (negative!).
- Dividing by a negative number FLIPS the inequality: $n > 10.32 \rightarrow$ at least 11.
- Always check: $0.8^{10} \approx 0.107$ (not enough), $0.8^{11} \approx 0.086$ (works).

3. Parent check method

Ask your child: “Is log 0.8 positive or negative? What happens to the inequality sign when you divide by it?”

Practice

1. A drug halves in the body every hour. After at least how many hours is less than 1% left?
2. Each pane of glass keeps 90% of the light. At least how many panes to get below 50%?
3. A substance decreases by 20% each year. After at least how many years is less than half left?
4. Find the smallest positive integer n with $(1/3)^n < 0.001$.
5. Light passes through 2 B-filters (each keeps 50%), then some A-filters (each keeps 80%). At least how many A-filters bring it below 1%?

Answers

1. 7 hours ($0.5^6 \approx 0.0156$, $0.5^7 \approx 0.0078$)
2. 7 panes ($0.9^6 \approx 0.531$, $0.9^7 \approx 0.478$)
3. 4 years ($0.8^3 = 0.512$, $0.8^4 \approx 0.410$)
4. $n = 7$ ($3^6 = 729$, $3^7 = 2187$)
5. 15 (need $0.8^n < 0.04$; $0.8^{14} \approx 0.044$, $0.8^{15} \approx 0.035$)

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